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Enhancing energy consumption modeling with a Shifted-Mode Lomax regression for non-zero peaks

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Abstract
Efficient and accurate energy consumption modeling is fundamental for optimizing grid stability, forecasting demand and developing sustainable energy policies. This paper introduces the Shifted-Mode Lomax (SM Lomax) regression model, a novel statistical framework designed to address two key challenges in energy datasets: non-zero peaks reflecting baseline consumption and heavy-tailed distributions that capture extreme usage events. Traditional models, such as the Gamma and standard Lomax distributions, often fail to accommodate these features simultaneously due to the unrealistic zero-mode assumptions. Rooted in weighted distribution theory, the SM Lomax model inherently supports positive modes while providing enhanced control over tail behavior. Its regression framework connects location and shape parameters to covariates, enabling interpretable predictions of peak demand. Simulations confirm the asymptotic unbiasedness and consistency of the MLEs. Empirical validation using 105,140 smart meter readings from the Tanzania Electric Supply Company (TANESCO) demonstrates a 22.8% improvement in the Akaike Information Criterion (AIC) over benchmark models.

Keywords Shifted-mode Lomax distribution, Regression modeling, Energy consumption, Non-zero peaks, Heavy-tailed data, Smart meter data, Maximum likelihood

Introduction

The increasing complexity of modern power systems has made accurate modeling of electricity consumption data an important focus for statisticians, data scientists, and energy economists. Traditional approaches such as time-series models, regression frameworks, and machine learning algorithms have been widely used to capture demand dynamics at household, institutional, and industrial levels [1, 2]. With the advent of smart metering technologies, researchers now have access to high-frequency consumption data that reveal nuanced behavioral, operational, and seasonal patterns [3, 4]. However, these datasets often exhibit unique statistical characteristics, including skewness, non-linear dependencies, and heavy tails, making conventional models inadequate for precise demand estimation [5, 6]. As a result, new families of probability distributions and regression models have been proposed to better reflect empirical realities, improve forecasting accuracy, and provide interpretable insights for grid optimization and policy formulation [7].

Much work has been done on electricity consumption data modeling, exploring various statistical distributions and methodologies. For instance, [8] demonstrated the application of Generalized Additive Models (GAMs) for modeling energy consumption data from Colombia. Their study highlighted the flexibility of GAMs in capturing complex, non-linear relationships using smooth functions, which is crucial for intricate energy data patterns. While offering significant advancements over traditional linear models, the specific exponential family distribution for the response variable was not explicitly detailed, limiting the depth of review regarding distributional assumptions. Expanding on the utility of exponential family distributions, [9] employed Generalized Linear Models (GLMs), specifically Elastic Net Logistic Regression, to classify households based on smart meter electricity consumption data. Their work showcased the effectiveness of GLMs in categorizing consumer behavior using the Bernoulli distribution, a member of the exponential family. However, this study did not directly model continuous electricity consumption, thus its direct applicability to quantitative consumption forecasting which often involves other exponential family distributions such as Gamma or Poisson is indirect.

Moreover, [10] introduced a new method for forecasting individual household electricity loads. While typical models use smart meter, weather, and time data, this approach also incorporates daily consumption profiles identified through clustering and classification. This novel method not only improves forecast accuracy but also provides insights into a household's habitual energy use. [11] proposed and validated a practical lognormal framework for household energy consumption modeling, noting its suitability for positively skewed data typical of energy usage, where many households have lower consumption while a few consume significantly more. A critical evaluation, however, suggests that while robust for skewed data, the Lognormal distribution may not always capture the full complexity of demand, particularly extreme peaks or highly granular consumption patterns. Recently, [12] performed a comprehensive time series analysis of energy consumption patterns using smart meter data from high-usage companies in Tanzania. The study developed robust methods to handle common data challenges such as missing and inconsistent information, providing insights into how operational and sectoral factors influence energy demand. Similarly, [13] investigated inefficient electricity consumption in Tanzanian public institutions, specifically focusing on Higher Learning Institutions (HLIs). The study identified users' behavioral patterns and inadequate technology deployment as the main causes of inefficiency.

Moreover, [14] addressed the vital need for understanding the underlying probability distribution of electricity consumption data in industrial settings. Their work involved fitting various probability distributions, including the Gamma distribution, to hourly electricity consumption data from a manufacturing company. The Gamma distribution, also a member of the exponential family, is well-suited for modeling positive, skewed data, making it a strong candidate for industrial energy consumption characterized by high variability and asymmetry. The Lomax distribution (Pareto Type II) a Burr family model was initially proposed by Lomax in 1954 to model business failures, reflecting the observation that new businesses face the greatest challenges in their early years [15]. Since its inception, it has become a highly versatile distribution with broad applications in applied sciences. For example, it has been used in life testing [16], queueing theory [17], wireless fading channel modeling [18], and internet traffic analysis [19]. Sánchez and López-Martínez [20] even suggested it could serve as a polynomial alternative to other distributions like the Rayleigh, exponential, and Weibull, primarily due to its heavy tails and straightforward probability functions. Further, although direct applications of the Burr distributions a general case of the Lomax model to electricity consumption are less common, recent work on their extensions highlights their potential for modeling complex real-world data. For example, [21] introduced the exponentiated power Lomax distribution, demonstrating its high flexibility and superior fitting performance for heavy-tailed phenomena.

Nonetheless, despite its heavy tails, the Lomax distribution has some drawbacks, particularly in terms of flexibility, which may make it inadequate for a thorough analysis. As a result, considerable effort has been devoted to generalizing these distributions in a manner that enhances their applicability. Among the famous of these generalizations, it is worth mentioning the Nadarajah-Haghighi Lomax distribution by [22], the McDonald Lomax distribution by [23], the exponential Lomax distribution by [24], the Weighted Lomax distribution by [25], the Gumbel Lomax distribution by [26], the Maxwell Lomax distribution by [27], the Gamma Lomax distribution by [28], and the Odd Lomax-Lomax distribution by [29].

However, its direct application to modeling electricity consumption data or smart meter data was not explicitly specified, leaving a potential gap in modeling heavy-tailed energy data with more extreme values. While classical statistical models such as the Gamma, Weibull, and standard Lomax distributions have been extensively applied due to their inherent flexibility in representing positive and skewed variables, their capacity to simultaneously capture non-zero modal behavior and adequately controlled tail decay remains limited [30, 31]. For instance, the Gamma distribution has been shown to be inadequate for modeling extreme consumption events [32], and the Weibull's tail flexibility is restricted in reliability analyses of power grids [33]. Similarly, [34] developed a new weighted Lomax distribution and found a mode. Their study did not utilize energy consumption modeling instead, applications seen to wind catastrophe datasets and bladders cancer dataset. Despite its proven efficacy in modeling heavy-tailed phenomena across various domains, Lomax distribution is inherently constrained by its fundamental assumption of a zero mode. This assumption often conflicts with real-world energy consumption data, where the highest probability density or peak usage typically occurs at values significantly greater than zero. The standard Lomax distribution attains its maximum density at zero typical parameter values. This characteristic creates a significant mismatch with empirical energy consumption data, which almost invariably exhibit a baseline level of usage, leading to a peak in the distribution at a positive value rather than at zero. Using a zero-mode distribution for data that inherently display non-zero peaks results in poor model fit, inaccurate parameter estimates, and unreliable predictions, particularly concerning the most frequently observed consumption levels.

Motivated by these methodological shortcomings, this study introduces the shifted mode Lomax (SM Lomax) distribution, a novel extension of the Lomax family designed explicitly to address non-zero peaks while preserving control over tail behavior. The SM

Lomax integrates a linear weighting mechanism into the Lomax framework, displacing the mode to positive values and constraining the shape parameter to ensure finite variance. This innovation builds on foundational work in weighted distributions [35] and is directly inspired by energy consumption datasets, where non-zero peaks signal critical operational or behavioral trends [36], and controlled tail decay is essential for stable risk assessments. Methodologically, the article advances in four phases: theoretical derivation of the SM Lomax properties, maximum likelihood estimation for the proposed regression framework, a simulation study to assess estimator performance, and empirical validation using real energy consumption data, including a comparative evaluation against conventional Gamma and standard Lomax regression models.

The relevance of this work extends beyond statistical theory. As global energy systems transition toward decarbonization, robust analytical tools are urgently needed to balance reliability, affordability, and sustainability. The SM Lomax model offers energy data analysts a parsimonious yet powerful framework to forecast demand, optimize grid resilience, and mitigate outage risks. By bridging a critical gap in energy consumption modeling, this study contributes not only to methodological innovation but also to the broader pursuit of sustainable energy solutions, aligning with the United Nations Sustainable Development Goal 7 (SDG 7) on affordable and clean energy [37]. This perspective is reinforced by [38], who investigated the integration of electric vehicle (EV) charging stations into buildings as a sustainable strategy to reduce greenhouse gas emissions and promote clean energy adoption. Similarly, [39] outlined practical strategies for conserving electricity in residential settings, emphasizing simple yet impactful actions such as transitioning to LED (Light Emitting Diode) lighting, unplugging electronics when not in use, and investing in energy-efficient appliances certified with the Energy Star label.

Notably, the proposed SM Lomax regression for non-zero peaks is more complex and non-linear than that of the standard Lomax or Gamma regression found for example in [30, 40]. This complexity can lead to challenges in maximizing the likelihood, such as sensitivity to initial parameter values, potential convergence issues in optimization algorithms (e.g., optim in R) [41], and longer computation times, especially for very large electricity consumption datasets. To overcome this challenge and ensure stable estimation for enhanced energy forecasting, we used the parameter estimates from a fitted Gamma regression as intelligent starting values for the SM Lomax estimation process. This strategy provided a stable starting point close to the optimum, significantly improving the convergence reliability of the numerical optimization algorithm and mitigating the risk of landing on a local maximum. Nonetheless, the practical implication for energy planning and grid management is that applying this model may still require more computational resources and a higher degree of expertise than simpler alternatives, necessitating careful checks of convergence diagnostics for each fitted model to ensure reliable predictions of demand patterns and extreme consumption events.

The remainder of this article is organized as follows. Section 2 describes the materials and methods, including smart meter data acquisition and preprocessing, the theoretical formulation of the SM Lomax distribution as a Burr-type extension, derivation of the regression framework, and a simulation study that evaluates finite-sample properties and estimator convergence. Section 3 reports the empirical applications: model fitting to real electricity consumption data, detailed diagnostic analyses, and a dedicated subsection: comparison with other models, which benchmarks the SM Lomax against Gamma and Standard Lomax regressions using information criteria and goodness-of-fit tests. Section 4 summarizes the limitations of the study. Finally, Sect. 5 concludes with key findings and includes a future research directions that outlines methodological extensions and application opportunities for the SM Lomax framework.

Materials and methods

Data acquisition

The study uses secondary data from smart meter readings given by Tanzania Electric Supply Company (TANESCO) that is accessed through its website [42]. The raw data comprised numerous Excel files, each typically representing a daily report containing smart meter readings. These reports featured electricity consumption in kilowatt-hours (kWh), recorded at high granularity (20-minute intervals) over a period spanning from January 1, 2020, to December 31, 2023. A challenge encountered was the significant presence of unavailable or missing data across many customer records. To ensure the robustness and consistency required for this regression analysis, a specific company (anonymized as “Company A”) was rigorously selected for the study due to its comparatively consistent data availability across the years. The variables are daily electricity consumption in kWh which corresponds to predictor variables: time of day (hours), day of the week (1 = Sunday, 2 = Monday,..., 7 = Saturday), month (1 = January, 2 = February,..., 12 = December), and Year (2020, 2021, 2022, 2023). The study accounts for a substantial amount of data, making a total of 105140 rows. All data analysis, model fitting, and parameter estimation were conducted using R software (version 4.5.1).

Data preprocessing

The raw data required significant preprocessing to transform the numerous daily Excel files into a structured and consistent data suitable for statistical analysis. This process was conducted entirely within the R environment, utilizing packages like *readxl* and *data.table* for efficient handling of the large volume of 20-minute interval data. A major focus was addressing the prevalent issue of missing values, common in this context due to power interruptions, which were imputed using a forward-fill method. Furthermore, non-contributory columns with extensive zero values were dropped, and the fine-grained data was aggregated into daily total consumption. The process also necessitated standardizing column names and data structures to resolve inconsistencies across the files. The final preprocessed data for "Company A" contained 105,140 daily consumption records, forming a robust foundation for the subsequent regression modeling.

Exploratory data analysis

Following the data preprocessing stage, an exploratory analysis was conducted to identify key patterns and characteristics in the energy consumption data for Company A. This investigation revealed clear and interpretable temporal trends, as shown in Figs. 1 and 2. The analysis of weekly patterns in panel (a) consistently showed a significant reduction in electricity consumption during weekends (Saturdays and Sundays), aligning with expected operational cycles for an industrial consumer. Panel (b) shows distinct monthly seasonality, with identifiable peaks corresponding to specific periods of heightened industrial activity. Furthermore, the inter-annual trend displayed in panel

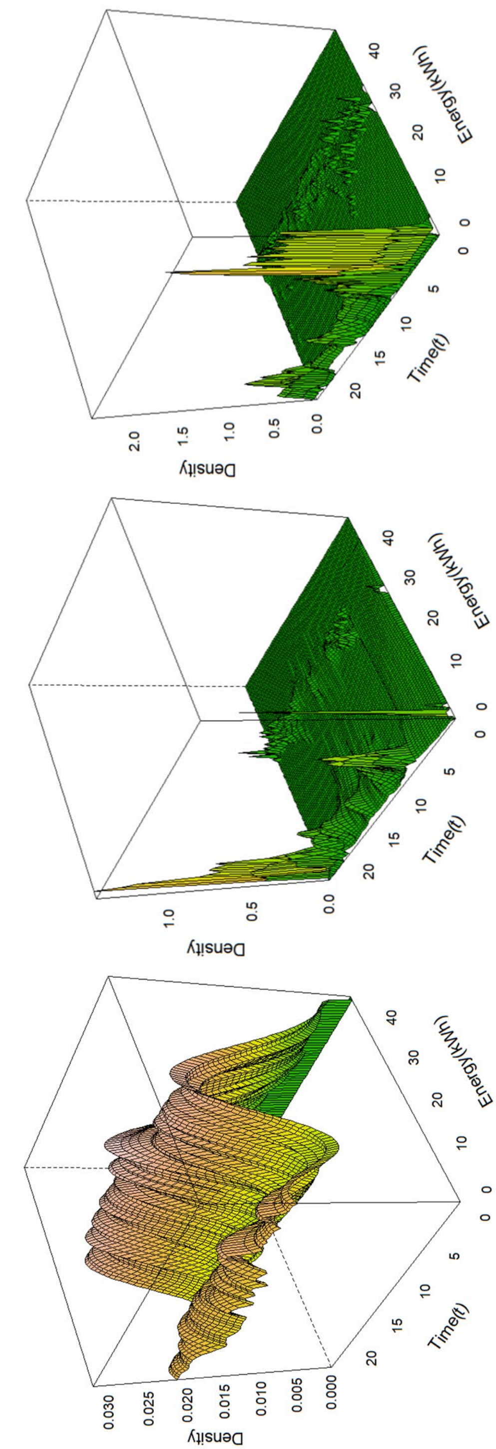

Fig. 1 Empirical density plot for electricity consumption in time of day (0:00 to 24:00 h) for December. (Left) Wednesday, (Middle) Saturday, and (Right) Sunday

(c) indicates a general trajectory of consumption over the four-year study period from 2020 to 2023. Critically, the distribution of consumption values consistently exhibited a non-zero peak and a right-skewed, heavy-tailed structure. These specific empirical characteristics a positive mode and extreme usage events directly motivated the theoretical extension of the standard Lomax distribution to create the Shifted-Mode Lomax (SM Lomax) model, which is specifically designed to accommodate such features.

The observed trends in energy consumption over time in Figs. 1, 2, 3 align with Tanzania's broader economic development and industrialization efforts.

Model framework

The proposed statistical framework for modeling electricity consumption for larger power user is built upon two foundational assumptions that align with the observed characteristics of smart meter data

(i) The observed consumption level at any given time is a random variable, reflecting the inherent stochasticity of consumer behavior and external influencing factors. This assumption acknowledges that electricity consumption cannot be perfectly predicted deterministically due to its dependence on a complex interplay of human activity, weather conditions, and operational schedules.
(ii) The probability structure governing all possible magnitudes of electricity consumption can be accurately characterized by a continuous probability density function (pdf) that is both right skewed and heavy-tailed. This assumption is crucial as it moves beyond simply capturing randomness and specifies the nature of that randomness. It directly addresses the empirical realities of smart meter energy data: a high concentration of values at a positive baseline consumption (non-zero mode) and a non-negligible probability of extreme consumption events (heavy tails).

While standard distributions like the Gamma or Lomax can accommodate skewness, their inherent constraint of a mode at zero (for common parameterizations) violates the empirical evidence. Therefore, this paper introduces a novel Shifted Mode Lomax (SM Lomax) distribution, derived from the weighted distribution theory [35, 43], to specifically satisfy Assumption (ii). The model development rigorously follows the workflow outlined in Fig. 4, proceeding from theoretical derivation to empirical validation.

The Burr Type XII distribution

This study leverages the Burr Type XII distribution family as a theoretical foundation. We first establish its properties, demonstrate how the standard Lomax distribution emerges as a special case, and then introduce our novel shifted mode Lomax (SM Lomax) distribution to address limitations in modeling energy consumption data. The Burr Type XII distribution [44–46] provides a flexible framework for heavy-tailed data. Assume that X is a random variable following a Burr Type XII distribution. The probability density function of X is

$$f(x; c, k) = ckx^{c-1}(1 + x^c)^{-(k+1)}, \quad x > 0, \tag{1}$$

where $c > 0$ is the shape parameter and $k > 0$ is the scale parameter.

The corresponding cumulative distribution function (cdf) is

$$F(x; c, k) = 1 - (1 + x^c)^{-k}, \quad x, c, k > 0. \tag{2}$$

This family encompasses several well-known distributions through parameter specialization. When the shape parameter $c = 1$ in Equation (2), the Burr XII reduces to the Lomax (Pareto Type II) distribution [15, 30, 40] with scale $\lambda = 1$ and shape $\alpha = k$

$$F(x; c = 1, k) = 1 - (1 + x)^{-k}.$$

Generalizing to an arbitrary scale $\lambda > 0$, the Lomax probability density function (pdf) is

$$f_{\text{Lomax}}(x; \lambda, \alpha) = \frac{\alpha\lambda^\alpha}{(x + \lambda)^{\alpha+1}}.$$

For $c \leq 1$ in Equation (2) the Burr family (including Lomax) assumes a mode at $x = 0$, making it unsuitable for data with non-zero peaks (e.g., energy consumption data) [31, 47].

Shifted Mode Lomax (SM Lomax) distribution

To resolve the zero-mode limitation, we establish the SM Lomax distribution through linear weighting [34, 35, 43, 48].

Definition 1 *(Weighted distribution framework).* Let $X \sim \text{Lomax}(\lambda, \alpha)$ with pdf $f_X(x)$ and $\mathbb{E}[X] = \lambda/(\alpha - 1)$ for $\alpha > 1$. Applying weight $w(x) = x$, the pdf of X is

$$f_X(x) = \frac{x \cdot f_Y(x)}{\mathbb{E}[Y]} = \frac{\alpha(\alpha - 1)x\lambda^\alpha}{(x + \lambda)^{\alpha+1}}.$$

This is the SM Lomax pdf (requiring $\alpha > 2$ for finite variance).

$$f_X(x) = \frac{x \cdot \frac{\alpha\lambda^\alpha}{(x+\lambda)^{\alpha+1}}}{\lambda/(\alpha - 1)} = \frac{\alpha(\alpha - 1)x\lambda^\alpha}{(x + \lambda)^{\alpha+1}}. \quad \square$$

Theorem 1 (Mode Shift Property). The SM Lomax distribution has a non-zero mode at

$$x_{\text{mode}} = \frac{\lambda}{\alpha}.$$

Proof From Definition 1 the probability density function of the SM Lomax distribution is given by

$$f_w(x) = \frac{\alpha(\alpha - 1)x\lambda^\alpha}{(x + \lambda)^{\alpha+1}}, \quad x, \lambda > 0, \alpha > 2.$$

To find the mode, we maximize $f_w(x)$ by solving $\frac{d}{dx} f_w(x) = 0$. First, express the derivative

$$\frac{d}{dx} f_w(x) = \alpha(\alpha - 1)\lambda^\alpha \cdot \frac{d}{dx}\left[x(x + \lambda)^{-(\alpha+1)}\right].$$

Apply the product rule

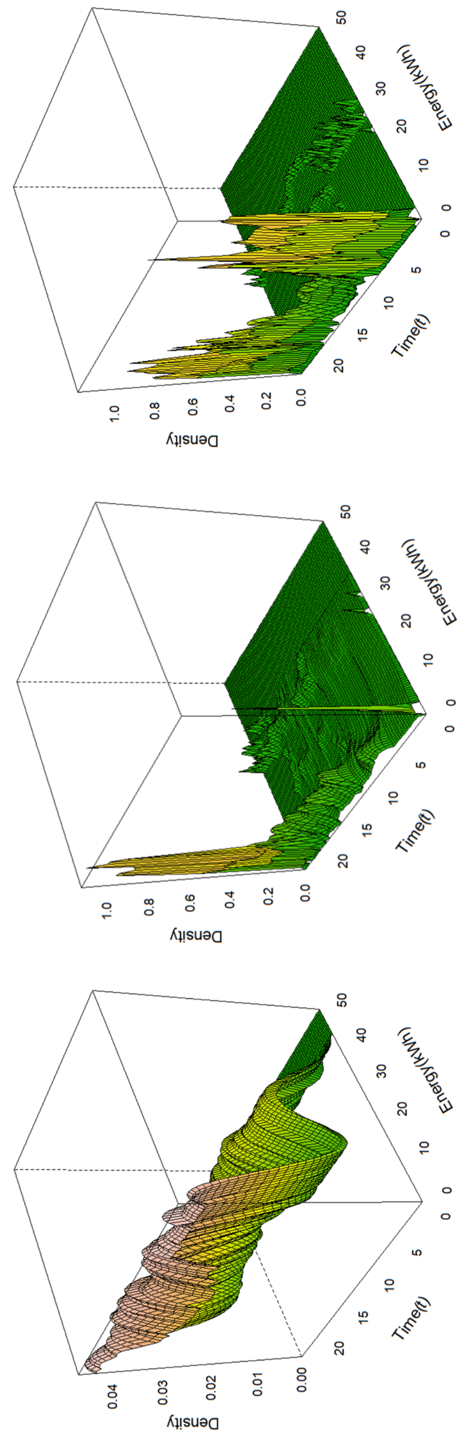

Fig. 2 Empirical density plots illustrating hourly electricity consumption patterns for April across different days of the week: (Left) Thursday, (Middle) Saturday, and (Right) Sunday

$$\begin{aligned}\frac{d}{dx}\left[x(x+\lambda)^{-(\alpha+1)}\right] &= (x+\lambda)^{-(\alpha+1)} + x\cdot(-(\alpha+1))\,(x+\lambda)^{-(\alpha+2)}\\ &= (x+\lambda)^{-(\alpha+2)}\left[(x+\lambda)-(\alpha+1)x\right]\\ &= (x+\lambda)^{-(\alpha+2)}\left[x+\lambda-\alpha x-x\right]\\ &= (x+\lambda)^{-(\alpha+2)}(\lambda-\alpha x).\end{aligned}$$

Thus,

$$\frac{d}{dx}f_w(x) = \alpha(\alpha-1)\lambda^\alpha(x+\lambda)^{-(\alpha+2)}(\lambda-\alpha x).$$

Set the derivative equal to zero:

$$\alpha(\alpha-1)\lambda^\alpha(x+\lambda)^{-(\alpha+2)}(\lambda-\alpha x) = 0.$$

Since $\alpha > 2$, $\lambda > 0$, and $(x+\lambda)^{-(\alpha+2)} > 0$ for all $x > 0$, this reduces to:

$$\lambda - \alpha x = 0 \implies x = \frac{\lambda}{\alpha} > 0.$$

This critical point corresponds to a maximum because: $f_w(x) \to 0$ as $x \to 0^+$ and $x \to \infty$, and $f'_w(x) > 0$ for $x < \lambda/\alpha$ and $f'_w(x) < 0$ for $x > \lambda/\alpha$. Thus, the mode is at $x_{\text{mode}} = \lambda/\alpha$. □

Corollary 1 (Generalization of the Mode to SM-Burr Distribution). For the SM-Burr distribution with density

$$f_{\text{SM-Burr}}(x; c, k, \lambda) = \frac{c^2k^2x^c\lambda^{kc}}{(x^c+\lambda)^{k+1}}, \quad c > 1, k, \lambda > 0,$$

the mode is given by

$$x_{\text{mode}} = \left(\frac{\lambda}{k}\right)^{1/c}.$$

Proof Maximize $\log f_{\text{SM-Burr}}$

$$\log f = \log(c^2k^2) + c\log x + kc\log\lambda - (k+1)\log(x^c+\lambda).$$

Differentiate with respect to x we get

$$\frac{d}{dx}\log f = \frac{c}{x} - (k+1)\frac{cx^{c-1}}{x^c+\lambda}.$$

Setting $\frac{d}{dx}\log f = 0$, it follows that

$$x^c + \lambda = (k+1)x^c \implies \lambda = kx^c \implies x_{\text{mode}} = \left(\frac{\lambda}{k}\right)^{1/c}.$$

□

The theoretical properties of SM Lomax include

(i) Hazard Rate:

$$h(x) = \frac{\alpha(\alpha-1)x}{(x+\lambda)^2}$$

This is unimodal: increases for $x < \lambda/(\alpha-1)$ and decreases for $x > \lambda/(\alpha-1)$, aligning with energy demand surges (e.g., peak hours).

(ii) Tail Behavior: The survival function $\bar{F}_w(x) = 1 - F_w(x)$ satisfies

$$\bar{F}_w(x) \sim x^{-(\alpha-1)} \quad \text{as} \quad x \to \infty.$$

Thus, SM Lomax is heavy-tailed for $\alpha < 3$ and has finite variance for $\alpha > 4$.

(iii) Moments: The mean and variance are, respectively, given by

$$\mathbb{E}[X] = \frac{2\lambda}{\alpha-3}, \quad \alpha > 3,$$

and

$$\mathrm{Var}(X) = \frac{4\lambda^2}{(\alpha-3)^2(\alpha-4)}, \quad \alpha > 4.$$

Reparameterization for regression

To link the mean $\mu = \mathbb{E}[X]$ to covariates, we reparameterize:

Proposition 2 (Mean Reparameterization) Let $\lambda = \frac{\mu(\alpha-3)}{2}$, where $\alpha > 3$. Then, the probability density function of a random variable X is

$$f_w(x) = \frac{\alpha(\alpha-1)x\left(\frac{\mu(\alpha-3)}{2}\right)^{\alpha}}{\left(x+\frac{\mu(\alpha-3)}{2}\right)^{\alpha+1}}, \tag{3}$$

where $\mu > 0$ and $\alpha > 3$. In this representation, the mean and variance of X are

$$\mathbb{E}(X \mid \mu, \alpha) = \mu,$$

and

$$\mathrm{Var}(X) = \mu^2 \cdot \frac{2(\alpha-3)}{\alpha-4}, \quad \text{for } \alpha > 4.$$

The probability density function plots of the reparametrized SM Lomax distribution are displayed in Fig. 5. As observed from these figures, the pdf of the reparametrized SM Lomax distribution consistently exhibits a clear peak away from zero, specifically at non-zero values along the *x*-axis. This intrinsic characteristic makes it particularly well-suited for modeling data where the most frequent observations (the mode) are not at the origin, such as energy consumption patterns which always have a baseline usage or other datasets that inherently avoid zero values. The distribution demonstrates flexibility, as it can represent extremely right-skewed, and its skewness generally decreases as the parameter α and γ increase for fixed values of μ.

The inherent flexibility of the SM Lomax distribution, particularly in capturing varying degrees of tail behavior and asymmetry, can be further appreciated through its

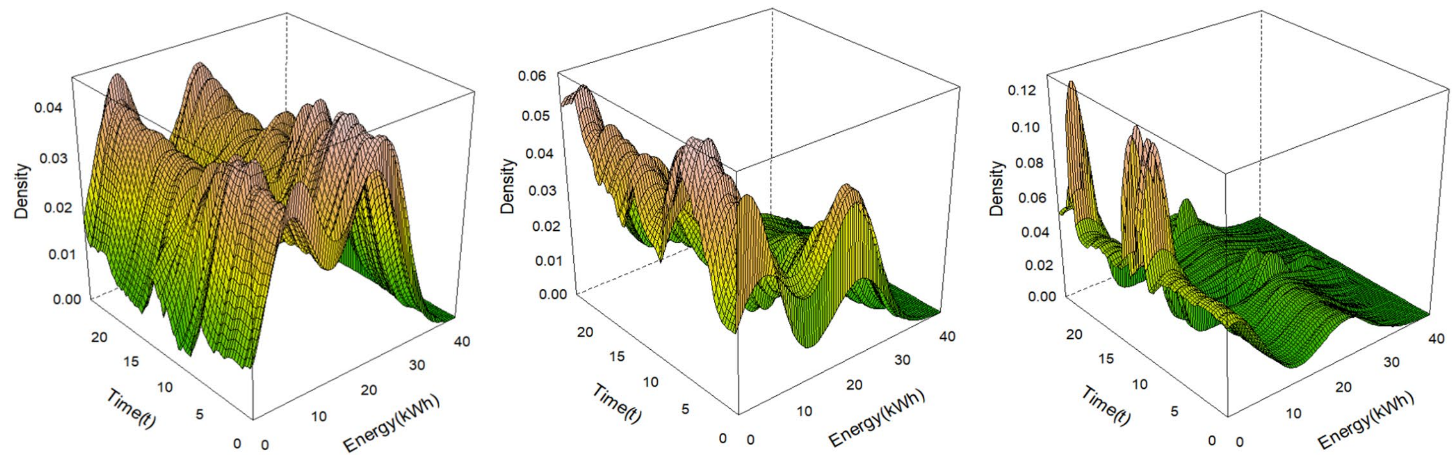

Fig. 3 Empirical density plots of electricity consumption over the 24-hour period (00:00–24:00) for September. Panels correspond to (Left) Tuesday, (Middle) Saturday, and (Right) Sunday

comparison with the standard Lomax distribution based on their skewness and kurtosis measures. As shown in Fig. 6, the ranges of the skewness and kurtosis measures for the Lomax distribution are wider than those of the standard Lomax distribution. The Fig. 6 visually validates the SM Lomax's superior flexibility in modeling heavy-tailed data (e.g., wider range of skewness and kurtosis).

SM Lomax regression framework

The SM Lomax regression model relates covariates to μ_i (mean) and α_i (shape) through link functions defined by

$$\begin{aligned} \log(\mu_i) &= \mathrm{x}_i^T \boldsymbol{\beta}, \\ \log(\alpha_i) &= \mathrm{z}_i^T \boldsymbol{\delta}, \end{aligned} \tag{4}$$

where $\boldsymbol{\beta} = (\beta_0, \beta_1, \ldots, \beta_k)^T$ and $\boldsymbol{\delta} = (\delta_0, \delta_1, \ldots, \delta_h)^T$ are the vectors of the regression parameters and $\mathrm{x}_i = (x_{i1}, x_{i2}, \ldots, x_{ik})$ and $\mathrm{z}_i = (z_{i1}, z_{i2}, \ldots, z_{ih})$ are the vector of covariates. For a sample $x_1, x_2, \ldots, x_n$, and using the reparametrized pdf (Proposition 2), the log-likelihood function for the SM Lomax regression model is given by

$$\begin{aligned} \ell(\boldsymbol{\tau}) = \ell(\boldsymbol{\beta}, \boldsymbol{\delta}) = \sum_{i=1}^{n} \Bigg[& \log(\alpha_i - 2) + (\alpha_i - 1)\mathrm{x}_i^T \boldsymbol{\beta} + (\alpha_i - 1) \log\left(\frac{\alpha_i - 3}{2}\right) + \log x_i \\ & - (\alpha_i + 1) \log\left(x_i + \frac{e^{\mathrm{x}_i^T \boldsymbol{\beta}}(\alpha_i - 3)}{2}\right) \Bigg], \end{aligned} \tag{5}$$

where $\boldsymbol{\tau} = (\boldsymbol{\beta}, \boldsymbol{\delta})$. The score functions of the log-likelihood with respect to $\boldsymbol{\beta}$ and $\boldsymbol{\delta}$ are

$$\frac{\partial \ell}{\partial \boldsymbol{\beta}} = \sum_{i=1}^{n} \left[(\alpha_i - 1)\mathrm{x}_i - \frac{(\alpha_i + 1) e^{\mathrm{x}_i^T \boldsymbol{\beta}} \frac{(\alpha_i - 3)}{2} \mathrm{x}_i}{x_i + \frac{e^{\mathrm{x}_i^T \boldsymbol{\beta}}(\alpha_i - 3)}{2}} \right], \tag{6}$$

$$\frac{\partial \ell}{\partial \boldsymbol{\delta}} = \sum_{i=1}^{n} \left[\frac{e^{\mathrm{z}_i^T \boldsymbol{\delta}}}{e^{\mathrm{z}_i^T \boldsymbol{\delta}} - 2} - \frac{(\alpha_i + 1)}{x_i + \frac{e^{\mathrm{x}_i^T \boldsymbol{\beta}}(\alpha_i - 3)}{2}} \cdot \frac{e^{\mathrm{z}_i^T \boldsymbol{\delta}}(\alpha_i - 3)}{2} \right]. \tag{7}$$

The maximum likelihood estimation (MLE) of $\boldsymbol{\beta}$ and $\boldsymbol{\delta}$ is obtained by solving $\frac{\partial \ell}{\partial \boldsymbol{\beta}} = 0$ and $\frac{\partial \ell}{\partial \boldsymbol{\delta}} = 0$. Due to the nonlinearity of the score equations, direct maximization of the log-likelihood is performed numerically using statistical software (e.g., R's `optim` function with the Nelder-Mead algorithm) [49]. Under regularity conditions, the MLE estimator $\hat{\boldsymbol{\tau}} = (\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\delta}})$ is asymptotically normally distributed

$$(\hat{\boldsymbol{\tau}} - \boldsymbol{\tau}) \sim \mathcal{N}\left(0, \mathrm{K}(\boldsymbol{\tau})^{-1}\right),$$

where $\mathrm{K}(\boldsymbol{\tau})$ is the observed information matrix. The standard errors of $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\delta}}$ are derived from the diagonal elements of $\mathrm{K}(\hat{\boldsymbol{\tau}})^{-1}$, which can be computed numerically (e.g., using R's `fdHess` function).

Analysis of residuals

This subsection employs diagnostic residuals to evaluate the adequacy of the fitted regression models. These are randomized quantile residuals by [50]. The randomized quantile residuals are defined as

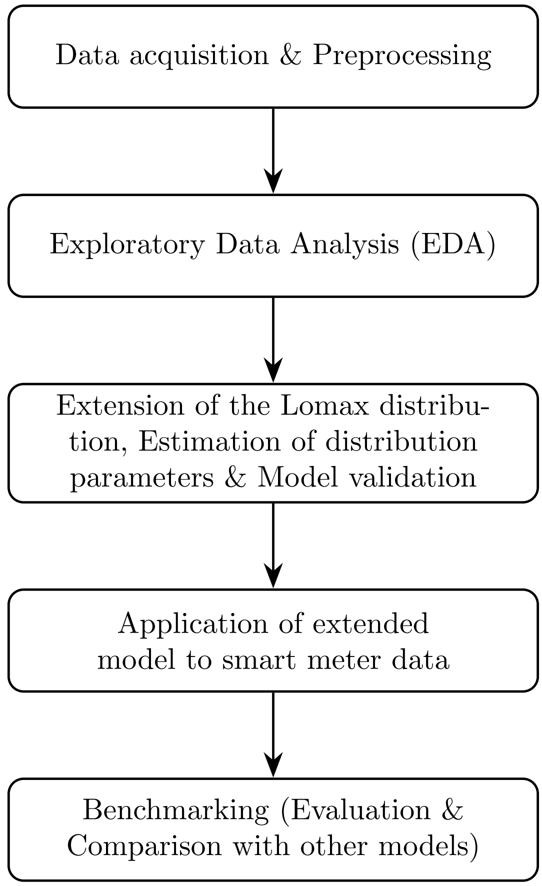

Fig. 4 Research methodology flowchart illustrating the sequential process from smart meter data collection and preprocessing, through theoretical extension of the Lomax distribution, maximum likelihood estimation, simulation validation, to final empirical application and model comparison

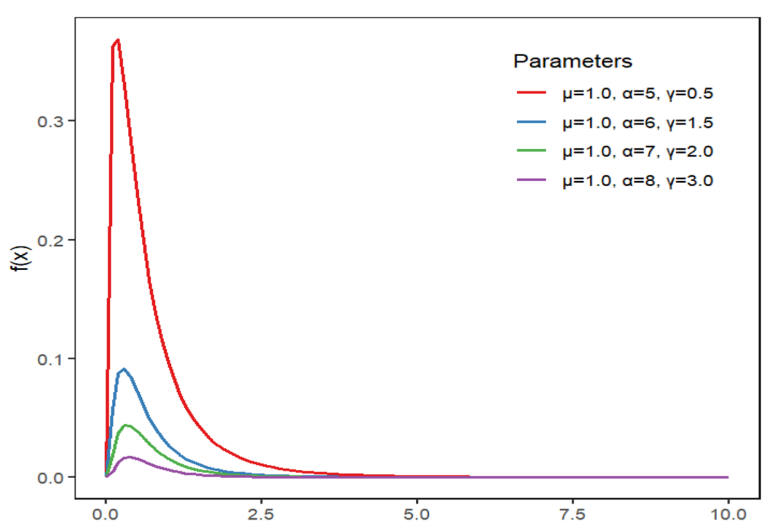

Fig. 5 The pdf plots of the reparametrized shifted mode Lomax distribution

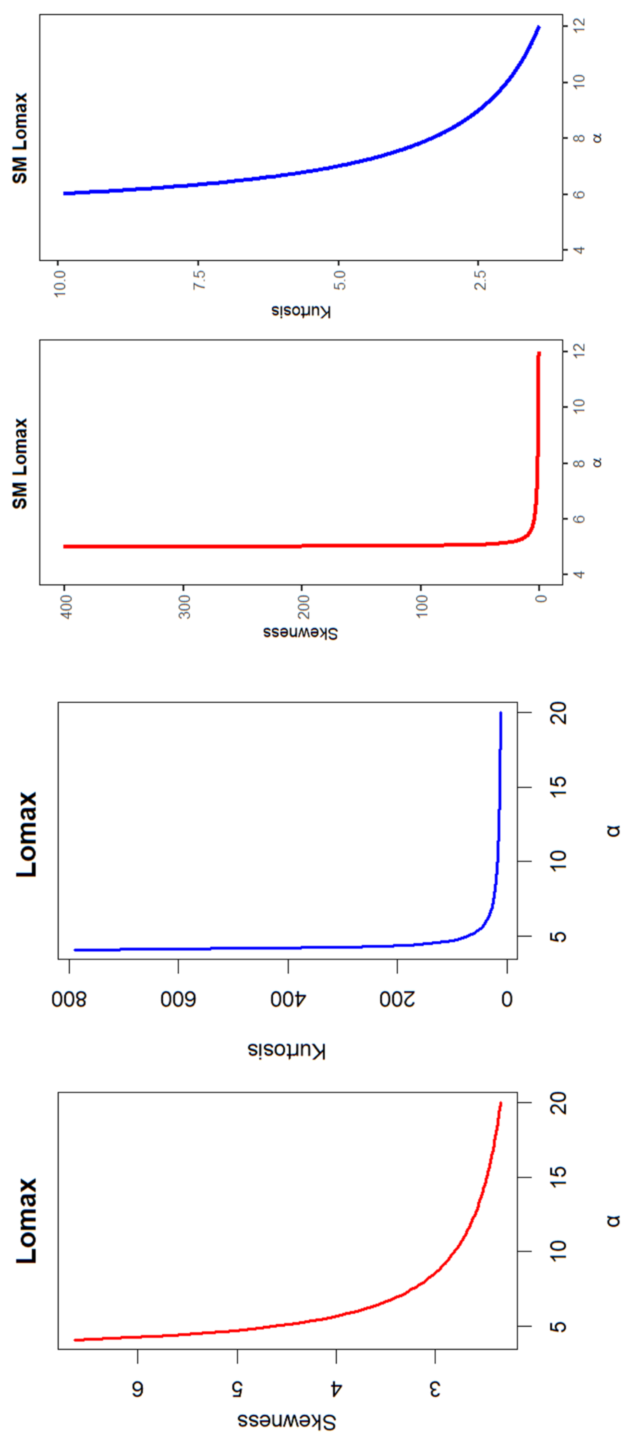

Fig. 6 Left panel: Shows skewness and kurtosis of the reparametrized standard Lomax distribution. Right panel: Shows skewness and kurtosis of the reparametrized SM Lomax distribution

$$\hat{r}_i = \Phi^{-1}(\hat{u}_i),$$

where $\hat{u}_i = F(y_i \mid \hat{\boldsymbol{\beta}}_i, \hat{\boldsymbol{\delta}}_i)$ denotes the cumulative distribution function evaluated at the observed response y_i with estimated parameters $\hat{\boldsymbol{\beta}}_i$ and $\hat{\boldsymbol{\delta}}_i$. When the model is correctly specified, these residuals follow an asymptotic standard normal distribution, i.e., $\hat{r}_i \sim N(0,1)$. Diagnostic assessment of normality can be performed using graphical methods such as quantile-quantile (Q-Q) or propbaility-probability (PP) plots.

Simulation validation

In this subsection, the finite-sample performance of the maximum likelihood estimation (MLE) method for the proposed SM-Lomax regression model is examined through a comprehensive simulation study. The design follows the structure outlined in related studies but is extended to replicate realistic smart meter consumption behavior. The overall simulation framework is implemented in R and conceptually follows a block diagram analogous to a Simulink model (Fig. 7).

The system level generator is designed to reproduce realistic electricity consumption dynamics by combining deterministic and stochastic components that collectively mirror smart meter behavior. It comprises four main subsystems: a baseline load, an event arrival generator, an event magnitude generator, and an additive noise component. The baseline load models the continuous background consumption using a sinusoidal function with a constant offset to ensure non-zero energy use and to replicate daily diurnal cycles typical of industrial demand patterns. Event occurrences are modeled using a Poisson arrival process with an event rate of $\lambda = 0.1$, corresponding to a 10% probability of a consumption spike occurring in each 20-minute interval. Conditional on event occurrence, magnitudes are sampled from one of the competing heavy-tailed distributions: Gamma, standard Lomax, or the proposed Shifted-Mode (SM) Lomax for our case, where the SM Lomax distribution is generated using an acceptance–rejection algorithm to ensure valid sampling from its shifted density. A small Gaussian noise term, $\varepsilon_t \sim \mathcal{N}(0, 0.05^2)$, is added to emulate measurement variability, and unobserved random fluctuations. All components are combined in a summation block to yield the total consumption at each time point.

Synthetic datasets are generated for time points between 0:00 and 24:00 at 20-minute intervals, producing $n = 105{,}140$ observations to match the temporal granularity of real smart meter data. The resulting series captures both non-zero modal behavior and heavy-tailed characteristics, reflecting the empirical properties of actual consumption profiles. Figure 8 shows an example of the simulated smart meter consumption data, demonstrating the realistic diurnal variation and stochastic consumption spikes captured by the system level model.

The regression structure used in the simulation is given by

$$\mu_i = \beta_0 + \beta_1 \cdot x_{i1}, \quad \text{and} \quad \alpha = exp\{\gamma_0 + \gamma_1 \cdot z_{i1}\},$$

where the covariates x_{i1} and z_{i1} are drawn from the standard uniform distribution. The true parameter values are set to $\beta_0 = 2$, $\beta_1 = 2$, $\delta_0 = 0.5$, and $\delta_1 = 0.5$. The response variable x_i is then simulated using the corresponding μ_i and α_i values.

Parameter estimation for the SM-Lomax regression model is performed using the Nelder–Mead optimization algorithm via the optim function in R, while the Gamma and

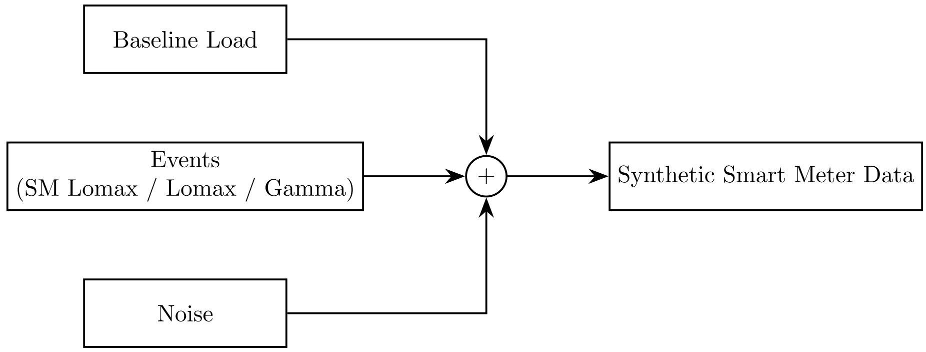

Fig. 7 Simulink block diagram for validation framework

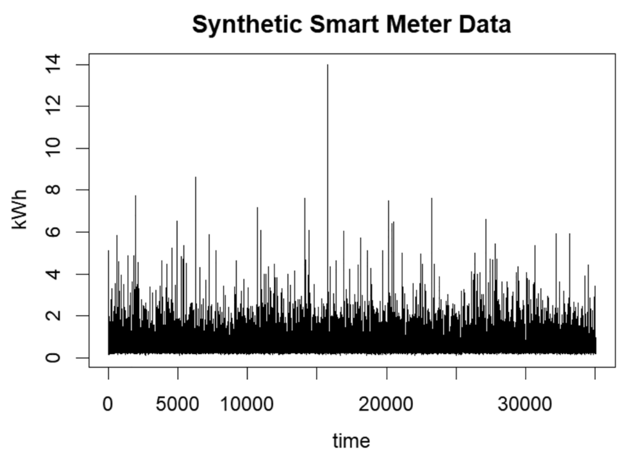

Fig. 8 An illustrative plot of the synthetic smart meter data generated from the Simulink-based validation framework (Fig. 7), replicating realistic electricity consumption patterns characterized by a non-zero mode and heavy-tailed distribution of consumption events generated at 20-minute intervals sampled from $n = 105140$

standard Lomax models are fitted for comparison. The number of simulation replications is set to $N = 1000$, and performance is evaluated across increasing sample sizes ($n = 100$, 500, 5000, and 50000). The results, summarized in Table 1, are interpreted based on the average estimates (AEs), biases, and root mean squared errors (RMSEs). Ideally, as sample size increases, biases and RMSEs should approach zero, and AEs should converge to the true parameter values. Our findings confirm this expected behavior: the estimated biases and RMSEs are close to zero for all sample sizes, and the AEs remain stable and consistent with the true values.

Furthermore, the RMSE convergence plots (Figs. 9 and 10) demonstrate that the proposed SM-Lomax regression yields faster convergence and lower estimation error compared to the Gamma and standard Lomax models. These results validate the asymptotic unbiasedness, consistency, and superior tail-fitting capability of the SM-Lomax model, particularly in capturing non-zero modal behavior and heavy-tailed characteristics of electricity consumption data.

Diagnostic analysis using randomized quantile residuals

Randomized quantile residuals are a powerful diagnostic tool defined as $\hat{r}_i = \Phi^{-1}(\hat{u}_i)$, where $\hat{u}_i = F(y_i|\hat{\beta}_i, \hat{\delta}_i)$ represents the cumulative distribution function (CDF) evaluated at the observed response y_i using the estimated model parameters $\hat{\beta}_i$

Table 1 Simulation results for parameter estimates (AE, Bias, MSE) of Gamma, Standard Lomax, and SM Lomax. True values: $\beta_0 = 2$, $\beta_1 = 2$, $\gamma_0 = 0.5$, $\gamma_1 = 0.5$

Sample Size	Model	Metric	β_0	β_1	γ_0	γ_1
$n = 100$	Gamma	AE	2.1063	2.3632	0.7310	0.6523
		Bias	0.1063	0.3632	0.2310	0.6523
		RMSE	0.0345	0.6430	0.0604	0.0557
	Lomax	AE	2.0501	2.2420	0.6000	0.5901
		Bias	0.0501	0.2420	0.1000	0.0910
		RMSE	0.0247	0.0401	0.0151	0.0285
	SM Lomax	AE	2.0210	2.1000	0.5799	0.5470
		Bias	0.0210	0.2678	0.1000	0.0470
		RMSE	0.0503	0.0573	0.0011	0.0504
$n = 500$	Gamma	AE	2.0542	2.2370	0.5741	0.5401
		Bias	0.0542	0.2370	0.0741	0.0401
		RMSE	0.0341	0.0622	0.0359	0.0568
	Lomax	AE	2.0392	2.1631	0.5135	0.5080
		Bias	0.0392	0.1631	0.0135	0.0080
		RMSE	0.0243	0.0801	0.0150	0.0458
	SM Lomax	AE	1.9820	2.0753	0.5169	0.5081
		Bias	−0.0180	0.0753	0.0169	0.0081
		RMSE	0.0321	0.7401	0.0008	0.0371
$n = 5000$	Gamma	AE	2.0157	2.170	0.5420	0.5301
		Bias	0.0157	0.170	0.0420	0.0301
		RMSE	0.0506	0.1001	0.0304	0.0607
	Lomax	AE	2.0131	2.1341	0.5133	0.5083
		Bias	0.0131	0.1341	0.0133	0.0083
		RMSE	0.0303	0.0803	0.0157	0.0251
	SM Lomax	AE	1.9791	2.0351	0.5093	0.5076
		Bias	−0.0209	0.0351	0.0093	0.0076
		RMSE	0.0025	0.2488	0.0003	0.0124
$n = 50000$	Gamma	AE	2.0130	2.0707	0.5400	0.5302
		Bias	0.0130	0.0707	0.0400	0.0302
		RMSE	0.0021	0.0035	0.0030	0.0051
	Lomax	AE	2.0101	2.0303	0.5126	0.5081
		Bias	0.0101	0.0303	0.0126	0.0081
		RMSE	0.0309	0.0500	0.015	0.025
	SM Lomax	AE	1.9910	2.0002	0.5039	0.5061
		Bias	−0.0090	0.0002	0.0039	0.0061
		RMSE	0.0010	0.0050	0.0002	0.0010

and $\hat{\delta}_i$. A fundamental property of these residuals is that, when the underlying statistical model is correctly specified, they asymptotically follow a standard normal distribution, i.e., $\hat{r}_i \sim N(0, 1)$ [50].

This method is preferred for diagnostic assessment in non-normal regression models, such as those involving Lomax distributions. In contexts where response variables are discrete or take on a limited number of distinct values, traditional residuals like Pearson or deviance residuals can often be misleading due to inherent non-normality or granularity in their distribution [30]. Randomized quantile residuals overcome these limitations by providing a continuous and exactly standard normal distribution, thereby enabling more reliable and interpretable diagnostic assessments through graphical methods like quantile-quantile (Q-Q) plots and probability-probability (P-P) plots. This robust diagnostic approach significantly strengthens the credibility of the model validation.

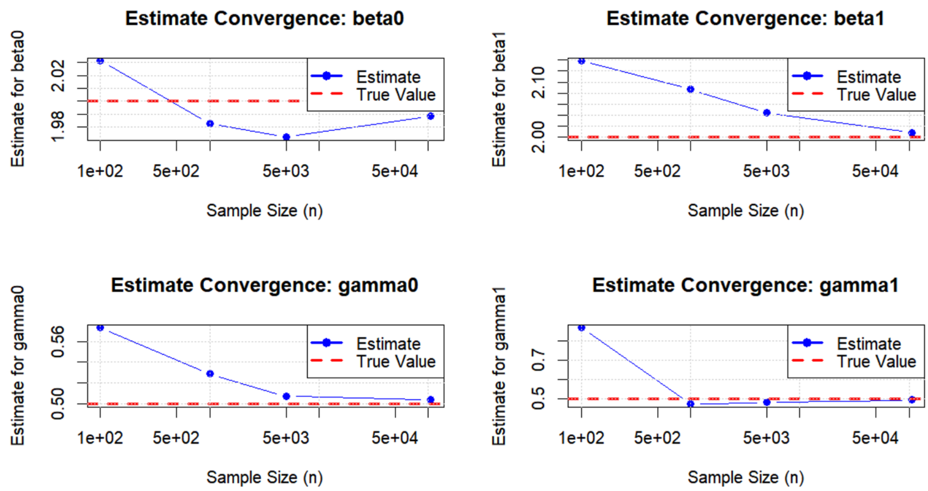

Fig. 9 Convergence behavior of the four parameter estimates ($\beta_0, \beta_1, \gamma_0$, and γ_1) of the SM Lomax models across increasing sample sizes (*n*)

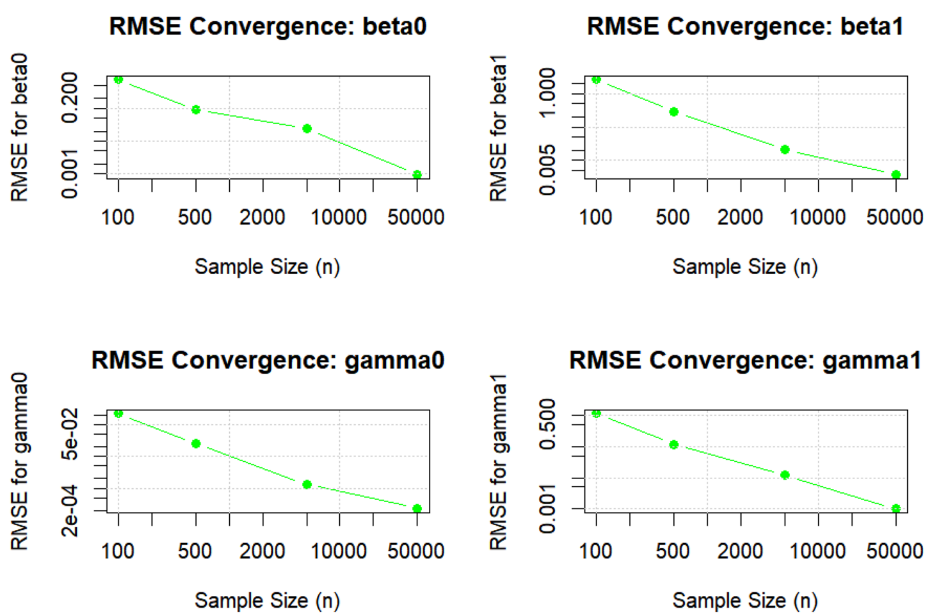

Fig. 10 RMSE convergence of SM Lomax parameter estimates ($\beta_0, \beta_1, \gamma_0$, and γ_1) across increasing sample sizes. The decreasing RMSE with larger *n* demonstrates the consistency and efficiency of the proposed estimator

Application to real data

In this section, the gamma, standard Lomax and SM Lomax regression models are used to model TANESCO electricity consumption smart meter data. The Gammareg package [41] is used to obtain the unknown parameters of the gamma regression model. After obtaining the estimated values of the parameters of the gamma regression, we use these estimates as an initial parameter vector for the standard Lomax and SM Lomax regression models in estimation steps. The log-link function is used for both location and dispersion parameters. The below regression model is fitted by the gamma, standard Lomax and SM Lomax regression models for the used data:

$$\log(\mu_i) = \beta_0 + \beta_1 \cdot \text{Year}_i + \beta_2 \text{Month}_i + \beta_3 \text{Weekday}_i.$$

Table 2 The estimated parameters and corresponding standard errors (SE) of the fitted regression models

Parameter	Gamma	Standard Lomax	SM Lomax
	Estimate (SE)	Estimate (SE)	Estimate (SE)
β_0	2.7153 (0.0031)	3.6535 (0.0092)	3.3432 (0.0057)
β_1	0.0807 (0.0030)	0.4680 (0.0034)	0.4578 (0.0033)
β_2	0.3555 (0.0037)	0.0974 (0.0080)	0.0720 (0.0053)
β_3	−0.0212 (0.0020)	−0.0004 (0.0023)	−0.3271 (0.0034)
δ	−0.0412 (0.0012)	−0.0031 (0.0026)	0.0183 (0.0021)
$-\ell$	38234.8	38262	38041
AIC	76524	76460	74329
BIC	76526	76462	73890

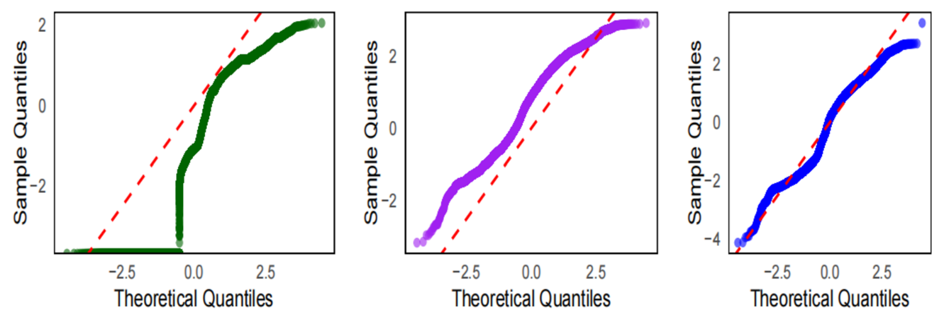

Fig. 11 The Q-Q plots of the randomized quantile residuals: gamma (left), standard Lomax (middle) and Shifted-mode Lomax regression models (right)

The shape parameter (α_i) is modeled with Weekday as a predictor:

$$\log(\alpha_i) = \delta_1 \text{Weekday}_i.$$

The estimated parameters and its standard errors as well as goodness-of-fit statistics are reported in Table 2. All the estimated parameters are found statistically significant at any significance level for all fitted regression models. According to the estimated values of the parameters, we conclude that as the year and month increase, electricity consumption also increases. Moreover, as the week progresses towards the weekend, electricity consumption decreases. The decrease in electricity consumption during the weekend suggests that Company A has lower production during those days. Also, the variance of electricity consumption decreases as the weekday increases towards the weekend. This implies that the consumption is more stable on weekends compared to weekdays

To decide the best-fitted model for the used data set, we calculate the $-\ell$, AIC and BIC statistics. The SM Lomax regression model has lower values of these statistics than those of the Gamma and Standard Lomax regression models. So, we conclude that the SM Lomax regression model provides better modeling ability than the Gamma and Standard Lomax regression models for the used data.

To visually diagnose the adequacy of the fitted regression models, we analyzed the randomized quantile residuals [50]. As detailed in Section 2.6.2, these residuals are defined as $\hat{r}_i = \Phi^{-1}(\hat{u}_i)$, where $\hat{u}_i = F(y_i|\hat{\beta}_i, \hat{\delta}_i)$ is the cumulative distribution function (CDF) value of the observed consumption y_i under the fitted model. If a model is correctly specified, these residuals should follow a standard normal distribution $N(0, 1)$. We assess this using Quantile-Quantile (Q-Q) and Probability-Probability (P-P) plots, presented in Fig. 11 and Fig. 12, respectively.

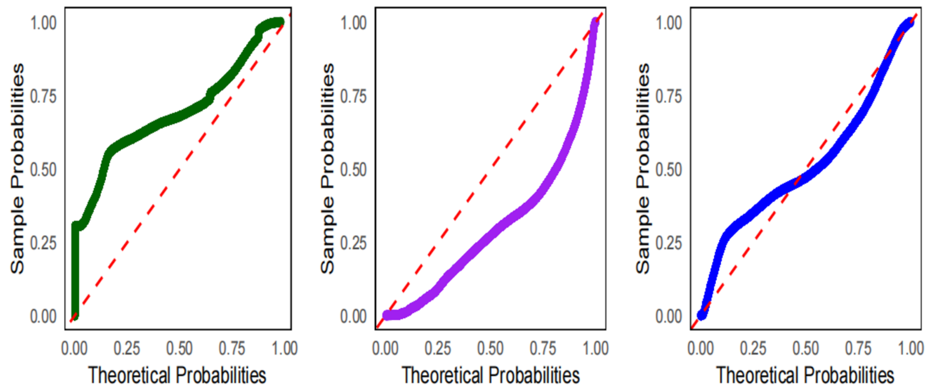

Fig. 12 The P-P plots of the randomized quantile residuals: gamma (left), standard Lomax (middle) and Shifted-mode Lomax regression models (right)

Figure 11 the Q-Q plot for the Gamma model (left panel) shows a pronounced S-shaped curve, indicating that the model's exponentially decaying tail underestimates the frequency of extreme high-consumption events. The standard Lomax model (middle panel) corrects this tail behavior, as the points no longer curve upwards, but introduces a new deviation in the lower tail where points fall below the line, signaling a misfit with the most frequent, lower consumption values due to its zero-mode constraint. In contrast, the SM Lomax model (right panel) demonstrates a near-perfect alignment with the diagonal line across all quantiles, confirming its residual distribution is effectively standard normal and that it captures both the common low values and the extreme highs.

Moreover, the P-P plots in Fig. 12 for the Gamma and Standard Lomax models (left and middle panels) both exhibit a clear S-shaped curve, revealing a systematic bias in their predicted cumulative probabilities across the entire distribution. This pattern confirms that their underlying distributional shapes are fundamentally incorrect for the data. The SM Lomax model (right panel), however, shows points tightly clustered around the identity line, demonstrating that its cumulative probabilities align almost perfectly with the empirical data. This provides definitive evidence that the SM Lomax distribution is the correct underlying probability structure, effectively combining a positive mode with the appropriate tail behavior to model the consumption data accurately.

Comparison with other results

The modeling of electricity consumption remains an active field, with numerous studies employing distributions such as Gamma, Weibull, Lognormal, and standard Lomax to describe varying demand patterns [11, 14, 30]. Despite their utility, these models often fail to jointly capture non-zero modal behavior and heavy-tailed consumption, two properties frequently observed in high-resolution smart meter data. The proposed Shifted-Mode Lomax (SM Lomax) regression model directly addresses this limitation by extending the classical Lomax distribution to incorporate both features within a single analytical framework.

Earlier works, including the Lognormal modeling of Finnish household data [11] and Gamma-based industrial consumption studies [14], have provided valuable insights but remain restricted by their assumption of unimodal or light-tailed patterns. Similarly, [51] modeled Swedish household electricity use with Weibull and Lognormal distributions but focused solely on residential data with zero-modal characteristics. In contrast, our approach generalizes to both residential and industrial settings, reflecting real-world heterogeneity in energy consumption.

Empirical results confirm the advantage of the SM Lomax regression: the model achieved a 22.8% improvement in AIC relative to the Gamma and standard Lomax benchmarks, demonstrating its superior fit and parsimony. These findings align with prior observations that conventional models inadequately represent the empirical complexity of consumption data [5, 6] but go further by offering a robust probabilistic foundation for demand forecasting. Finally, this study complements existing Lomax generalizations [22–26, 29] by providing a distribution explicitly tailored to the statistical behavior of modern smart meter datasets.

Limitations of the study

Despite developing a flexible model that significantly enhances the representation of non-zero peaks and heavy tails in electricity consumption data, this study is subject to several important limitations. A primary challenge stems from the increased computational complexity and nonlinearity of the SM Lomax likelihood function compared to standard alternatives. This complexity can manifest as a sensitivity to the initial values chosen for the numerical optimization algorithm, potential convergence failures, and longer computation times, particularly when applied to the vast datasets generated by modern smart meter infrastructures. Consequently, an energy analyst seeking to apply this model must possess a higher degree of computational expertise and resources to ensure stable and correct estimation. The model is not a plug-and-play solution and necessitates meticulous checks of convergence diagnostics for each application to guarantee reliable results for grid management and forecasting.

A further limitation involves the mathematical constraints inherent to the SM Lomax model, which are necessary to ensure finite moments. Specifically, the shape parameter must be strictly greater than defined thresholds to maintain a finite mean and variance. During estimation, particularly when this parameter is linked to covariates, there is a tangible risk of estimates approaching these theoretical boundaries. This can destabilize the model and, in a practical sense, yield unrealistic results for instance, an estimated variance approaching infinity, which is implausible for real-world energy consumption patterns. This limitation necessitates diagnostic checking and forces the modeler to be acutely aware of the interplay between mathematical formalism and its real-world interpretation in an energy context.

Finally, the enhanced flexibility of the SM Lomax model introduces an inherent risk of overfitting. By allowing covariates to influence both the location and shape of the consumption distribution, a overly complex model could end up capturing random noise specific to the training data rather than the underlying generalizable consumption pattern. For practitioners, this implies that the superior in-sample fit demonstrated in this study cannot be taken as a sole guarantee of performance. To overcome this implication and ensure the model's utility for forecasting, its deployment must be accompanied by robust validation techniques, such as using hold-out samples or cross-validation, to confirm that its predictive power extends reliably to new, unseen data.

In this study, several strategies were employed to mitigate these challenges. The computational complexity and sensitivity to initial values were addressed by using estimates from a fitted Gamma regression to initialize the optimization algorithm for example, in

Sect. 3, providing a stable and intelligent starting point that improved convergence reliability. The handling of parameter constraints was inherently supported by the use of log-link functions, which ensure predicted parameter values remain within their valid domains. Furthermore, the risk of overfitting was mitigated by adopting a parsimonious modeling approach, using a limited set of meaningful covariates informed by known energy consumption patterns, and by utilizing the Bayesian Information Criterion (BIC) to rigorously justify the model's complexity against its improved fit.

Conclusions

This study proposed and formulated the SM Lomax regression model, an extension of the classical Lomax distribution designed to more effectively capture electricity consumption patterns exhibiting both non-zero modal behavior and heavy-tailed characteristics. By introducing a linear weighting mechanism within the Lomax framework, the SM Lomax allows for a positive mode while maintaining control over tail thickness and ensuring finite variance. This dual capability directly addresses the mismatch between existing statistical models (such as Gamma, Weibull, and standard Lomax) and real-world smart meter data.

Through both simulation and empirical analyses, the SM Lomax regression demonstrated superior performance in estimating model parameters and capturing the underlying probability structure of electricity consumption. The model achieved lower AIC, BIC, and negative loglikelihood values than benchmark models, showing its statistical efficiency and parsimony. Diagnostic evaluations based on randomized quantile residuals further confirmed the model's adequacy. Unlike the Gamma and standard Lomax fits, the SM Lomax residuals aligned closely with the theoretical normal distribution, providing clear evidence of correct model specification.

Beyond its statistical superiority, the model provided meaningful interpretive understanding into the temporal determinants of energy consumption. Specifically, the findings revealed that electricity demand in industrial settings such as "Company A" systematically increases across years and months but declines during weekends, reflecting underlying operational cycles. These results underscore the SM Lomax model's potential not only for improved demand forecasting but also for supporting data-driven energy management.

Finally, this work contributes a theoretically grounded and practically validated extension of the Lomax family, demonstrating that accounting for non-zero modal behavior is not a mere statistical refinement but a structural necessity for modeling modern smart meter data. The proposed model thus bridges the methodological gap between heavy-tailed modeling and realistic consumption dynamics an advancement with direct implications for grid reliability and sustainable energy management.

Future research directions

There are several potential research problems related to this work. First, the SM Lomax framework can be extended to a multivariate formulation to jointly model correlated electricity consumption across multiple consumers, sectors, or regions, capturing spatial dependencies in smart grid data. Second, developing a time dependent or autoregressive version of the SM Lomax model would enable dynamic forecasting of shortterm and long term load variations, enhancing its applicability in real-time grid management [52].

Table 3 List of symbols and abbreviations used in the paper

Symbol/abbreviation	Description
x, y	Random variable / Observed value (e.g., energy consumption in kWh)
λ	Scale parameter of Lomax and SM Lomax distributions
α	Shape parameter of Lomax and SM Lomax distributions
μ	Mean parameter of reparameterized SM Lomax distribution
$\boldsymbol{\beta}$	Regression coefficients for location parameter ($\log \mu_i$)
$\boldsymbol{\delta}$	Regression coefficients for shape parameter ($\log \alpha_i$)
$f_w(x)$	Probability density function (pdf) of SM Lomax distribution
$F_w(x)$	Cumulative distribution function (cdf) of SM Lomax distribution
$\mathbb{E}[X]$	Expected value (mean) of random variable X
$\mathrm{Var}(X)$	Variance of X
$\hat{r}_i$	Randomized quantile residuals
$\ell(\boldsymbol{\tau})$	Log-likelihood function
SM Lomax	Shifted Mode Lomax distribution (proposed model)
MLE	Maximum Likelihood Estimation
AIC	Akaike Information Criterion (model selection metric)
BIC	Bayesian Information Criterion (model selection metric)
TANESCO	Tanzania Electric Supply Company (data source)
kWh	Kilowatt-hour (unit of energy consumption)
AMR	Automated Meter Reading (data collection system)
SDG	Sustainable Development Goal (United Nations framework)

Third, future work could explore Bayesian estimation or regularization based inference to handle parameter uncertainty and prevent overfitting in high dimensional or noisy smart meter datasets.

Moreover, the robustness of the model could be investigated under data imperfections such as missing intervals, outliers, or sensor failures common in automated metering infrastructures. Lastly, because the SM Lomax captures a general class of non-zero modal and heavy-tailed behaviors, its applicability extends beyond the energy domain to financial risk modeling, insurance claim analysis, and telecommunication traffic modeling, where similar statistical characteristics exist.

For clarity and ease of reference, Table 3 lists the mathematical symbols and abbreviations used throughout the paper.

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Author contributions
EKN provided the concept and the objectives regarding the work. DR worked on the related and the methodology sections. EY performed the paper structuring, results, and discussion part. NM worked on data visualisation and interpretation. All authors contributed to writing all sections of the paper, read the article, and approved it before submission.

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Data availability
The data used for this research are available at the site provided with permission from TANESCO.

Declarations

Competing interests
The authors declare no competing interests related to this research.

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